TAUT CONTINUOUS HOMOMORPHISMS PRESERVE BEING WEAKLY PRE-ADIC

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Tautness of a continuous homomorphism of linearly topologized rings. Same chapter, Definition definition-taut. Gives openness and the fundamental-system property for the closures of the I_n B.
- Weakly pre-adic topological rings. Same chapter, Definition definition-weakly-adic. Fixes what has to be produced for B: an ideal whose power closures are open and form a fundamental system.

Lemma 0.1. *Let $\varphi : A \to B$ be a continuous homomorphism of linearly topologized rings. If φ is taut and A is weakly pre-adic, then B is weakly pre-adic.*

Proof. Let $I \subset A$ be an ideal such that the closure I_n of I^n is open and these closures define a fundamental system of open ideals. Then the closure of $I^n B$ is equal to the closure of $I_n B$. Since φ is taut, these closures are open and form a fundamental system of open ideals of B. Hence B is weakly pre-adic. □

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